OUTPUT

```
Windows PowerShell 5.1
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PS C:\Users\jamun> ollama serve
Error: listen tcp 127.0.0.1:11434: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.
PS C:\Users\jamun> ollama run llama3.2
>>> Explain artificial intelligence in simple words
Artificial Intelligence (AI) is a way to create machines that can think and learn like humans. Here's a simplified
explanation:

**What is AI?**

AI is a computer system that can:

1. **Learn**: AI systems can gather information from data and use it to improve their performance.
2. **Make decisions**: AI can analyze data and make choices based on that data.
3. **Solve problems**: AI can identify patterns and solve problems on its own.

**How does AI work?**

AI works by using complex algorithms (like recipes for computers) to analyze data and make decisions. These
algorithms are designed by humans, but once they're created, the AI system can run them on its own.

**Types of AI**

There are three main types of AI:

1. **Narrow AI** (also known as Weak AI): This type of AI is designed to perform a specific task, like recognizing
faces or playing chess.
2. **General AI** (also known as Strong AI): This type of AI is designed to perform any intellectual task that a
human can.
3. **Superintelligence**: This type of AI is significantly more intelligent than humans and could potentially
solve complex problems.

**Real-life examples of AI**

1. Virtual assistants (like Siri, Alexa, or Google Assistant)
2. Self-driving cars
3. Image recognition systems (like Facebook's facial recognition)
4. Chatbots and customer service systems
```

```
**Examples of AI:**

1. **Virtual assistants**: Siri, Alexa, and Google Assistant.
2. **Image recognition**: Facebook's photo tagging feature.
3. **Self-driving cars**: Cars that can navigate and make decisions on their own.
4. **Recommendation systems**: Netflix and Amazon's personalized suggestions.

**Why AI is important:**

1. **Efficiency**: AI can automate tasks, freeing up time for humans.
2. **Innovation**: AI can help us create new products and services.
3. **Improving lives**: AI can help us with healthcare, education, and many other areas.

That's a basic overview of artificial intelligence!
PS C:\Users\jamun\OneDrive\Desktop\Gen Ai\week - 2> python -m unittest test_client.py
.
----------------------------------------------------------------------
Ran 1 test in 0.000s

OK
```

```
PS C:\Users\jamun\OneDrive\Desktop\Gen Ai\week - 2> python main.py
Choose provider (gemini/ollama): ollama
Enter your question: Explain artificial intelligence in simple words

LLM Response:
Artificial intelligence (AI) is a way to create machines that can think and learn like humans. Here's how it works:

**How it works:**

1. **Data**: We feed a machine with lots of information, like pictures, words, and numbers.
2. **Algorithms**: We give the machine a set of rules to follow, like a recipe.
3. **Processing**: The machine uses these rules to analyze the data and make decisions.
4. **Learning**: The machine gets better at making decisions as it sees more data and experiences.

**What AI can do:**

1. **Recognize patterns**: AI can spot patterns in data, like recognizing a cat in a picture.
2. **Make decisions**: AI can make choices, like suggesting products to buy.
3. **Talk to humans**: AI can understand and respond to human language, like chatbots.
4. **Improve over time**: AI can get better at its tasks as it learns from more data.

**Examples of AI:**

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**Why AI is important:**

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```

```
PS C:\Users\jamun\OneDrive\Desktop\Gen Ai> cd "week - 2"
PS C:\Users\jamun\OneDrive\Desktop\Gen Ai\week - 2> python -m pip install -r requirements.txt
Requirement already satisfied: python-dotenv in C:\Users\jamun\AppData\Local\Programs\Python\Python313\Lib\site-packages (from -r requirements.txt (line 2)) (1.2.4)
Collecting pytest (from -r requirements.txt (line 3))
  Downloading pytest-9.2.0-py3-none-any.whl.metadata (7.9 kB)
Collecting google-genai (from -r requirements.txt (line 4))
  Downloading google_genai-3.3.0-py3-none-any.whl.metadata (61 kB)
Collecting requests (from -r requirements.txt (line 5))
  Downloading requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Collecting colorama>=0.4 (from pytest->-r requirements.txt (line 3))
  Using cached colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Collecting iniconfig>=2 (from pytest->-r requirements.txt (line 3))
  Downloading iniconfig-2.3.1-py3-none-any.whl.metadata (2.8 kB)
Collecting packaging>=24 (from pytest->-r requirements.txt (line 3))
  Downloading packaging-26.3-py3-none-any.whl.metadata (3.5 kB)
Collecting pluggy<2,>=1.6 (from pytest->-r requirements.txt (line 3))
  Downloading pluggy-1.7.0-py3-none-any.whl.metadata (5.0 kB)
Collecting pygments>=2.15 (from pytest->-r requirements.txt (line 3))
  Downloading pygments-2.21.0-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: anyio<5.0.0,>=4.8.0 in C:\Users\jamun\AppData\Local\Programs\Python\Python313\Lib\site-packages (from google-genai->-r requirements.txt (line 4))
(4.16.0)
Collecting google-auth<3.0.0,>=2.56.0 (from google-auth[requests]<3.0.0,>=2.56.0->google-genai->-r requirements.txt (line 4))
  Downloading google_auth-2.62.2-py3-none-any.whl.metadata (6.0 kB)
Requirement already satisfied: httpx<3.0.0,>=2.0.0 in C:\Users\jamun\AppData\Local\Programs\Python\Python313\Lib\site-packages (from google-genai->-r requirements.txt (line 4))
 (2.16.0)
Requirement already satisfied: pydantic<3.0.0,>=2.12.5 in C:\Users\jamun\AppData\Local\Programs\Python\Python313\Lib\site-packages (from google-genai->-r requirements.txt (line
4)) (2.14.3)
Collecting tenacity<9.2.0,>=8.2.3 (from google-genai->-r requirements.txt (line 4))
  Downloading tenacity-9.1.4-py3-none-any.whl.metadata (1.2 kB)
Collecting websockets<18.0,>=13.0.0 (from google-genai->-r requirements.txt (line 4))
```

Creating environment...

```
**The future of AI**

As AI continues to evolve, we can expect to see more machines that can:

1. **Improve themselves**: AI systems that can learn and adapt on their own.
2. **Interact with humans**: AI systems that can understand and respond to human emotions and needs.
3. **Solve complex problems**: AI systems that can tackle complex challenges in fields like healthcare, finance
and sustainability.

That's a basic overview of AI!

>>> Send a message (/? for help)
```